Ezequiel Boero <ezequiel.boero@adagioinstitute.org>

Re: Convex Perturbation Paper - My Response to Your Latest Comments

Ezequiel Boero <ezequiel.boero@adagioinstitute.org> Sun, Sep 27, 2026 at 8:15 PM
To: "Benjamin D. Summers" <bsummers@adagiollc.com>
Cc: "ISMAIL-SUTTON, SARA R." <sara.r.ismail-sutton@durham.ac.uk>

Hi Ben,

All four wording points and both results are in.
One amendment remains but overall I would consider the review of this paper closed with essentially nothing outstanding.

What checks out. Proposition 4.10 I re-derived in both directions; it is right. Remark 4.5's attainment condition is correct as a conditional statement.
The four wording points are in as described.

The re-entry point at ε = 0.01 is 0.4882, not the 0.4885 I reported. I had computed it with an M* taken from an unresolved grid of triples.
The manuscript prints the right figure.

The closed forms improve on what I gave you, which was numerical throughout, and you closed something I had explicitly declined to claim:
the rebound peak is exactly 1/64, and exactly so for every g*, because the $(1 - g^*)^2$ cancels against $M^* = 1/[16(1 - g^*)^2]$.
I had noticed the coincidence and left it as probably accidental. And on the null column in Remark 6.2 you supplied the reason
(both one-time laws summing to one forces a state on each side of g).
Shorter than the route I checked, which was that $\Sigma_j m_s(j)w_s(j) = g$ exhibits g as a weighted average of the Bayes weights.

The suggested amendment. The optimality claim in the abstract (that no bound in D_CK and M* alone improves Theorem 4.3) is a
statement at the scale level, since that is where Theorem 4.3 quantifies. The witness I supplied for it is the λ = 0.35 configuration,
which lives in the Section 4.3 example, and Section 4.3 says of itself that it "does not construct a continuous two-time family, and the
scale-level suprema are not realized." So what is established is the optimality of the pointwise problem.
The conditional statement in Remark 4.5 is untouched; what is missing is a witness of the right kind.

The repair is short, and it is your own Proposition 4.10 read on the other side of zero.
Take two distinct divisible families Γ_a and Γ_ref and h > 0, and set

$$\Gamma_0 = (\Gamma_a + h\Gamma_{ref})/(1 + h),$$

a convex combination of divisible families, hence a valid family and generically indivisible.
Identity (6) is algebraic in g, so evaluating at g = −h, where the affine path returns Γ_a, gives 0 = (1 + h)(X + hK) and therefore X = −hK at every triple. Hence $D_{CK}(\delta t;\Gamma_0) = h\, M^*(\delta t)$ and, for every $g \in [0,1]$,

$$D_{CK}(\delta t;\Gamma_g) = (1 - g)(h + g)\, M^*(\delta t) = (1 - g)[D_{CK}(\delta t;\Gamma_0) + g\, M^*(\delta t)] = E(g),$$

identically and at family level, so $g_{first} = g_{term} = g_1$ exactly.
I checked it on 1,140 triples with λ_a = 1, λ_ref = 2, h = 0.6: $\max\|X + hK\| = 3.9\cdot10^{-16}$ and max over [0,1] of $|D_{CK}(\Gamma_g) - E(g)| = 5.7\cdot10^{-17}$.

What makes this worth stating rather than merely inserting: it is Proposition 4.10's formula, $\Gamma_0 = (\Gamma_a - g^*\Gamma_{ref})/(1 - g^*)$, at $g^* = -h$.
The two expressions are identical, and the entrywise condition $\Gamma_a \geq g^*\Gamma_{ref}$ is automatic once $g^* < 0$, since $\Gamma_a \geq 0 \geq g^*\Gamma_{ref}$
(which is why the construction needs no hypothesis). The interior zero and the attainment of the envelope are then one construction with
the divisible family placed on either side of Γ_0 along the affine line: $g^* \in (0,1)$ gives $X = g^*K$, cancellation, an interior zero and separated
thresholds; $g^* < 0$ gives $X = -hK$, anti-alignment, and the envelope attained.

Two sentences appended to Proposition 4.10 would carry it, and the proposition would then account for both phenomena instead of one.

If you would rather not extend the proposition, the alternative is to weaken the abstract to the pointwise optimality, which the λ = 0.35
configuration does establish.

One trivial rider while you are in Remark 6.2: the statement that the weight reduces to the constant g exactly when the marginals
coincide wants "for 0 < g < 1", since at the endpoints it is constant regardless.

In summary, the substantive questions of both reports are answered and I have nothing further to ask on this paper.

Best,
Ezequiel

On Thu, Sep 24, 2026 at 2:45 AM Benjamin D. Summers <bsummers@adagiollc.com> wrote:

> Hi Ezequiel and Sara,
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> Sara, noted that you have no further points; thank you.
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> Ezequiel, thank you for the verification record and for the two results; I adopted both, and the four wording points. The revised manuscript is on SSRN under the same DOI (https://dx.doi.org/10.2139/ssrn.6812461):
>
> Attainment (your item 1). Remark 4.5 now states that if a triple attains both suprema with K = −cX and c = M_/D_CK, then S(g) = D_CK + gM_, the deficit equals E(g) on [0,1], and g_first = g_term = g_1; hence no bound in D_CK and M_* alone improves Theorem 4.3. Section 4.3 realizes the configuration: K = 2(q − 0.3)(0.2 − q)Z with q = ½(1 − e^{−2λ}), aligned against X for λ ∈ (0.2554, 0.4581); at λ = 0.35 the response equals E to machine precision and g_first = g_term = g_1 = 0.5311 at ε = 0.04, against g_0 = 0.5555. The abstract and the contributions paragraph state the attainment.
>
> Scale-level family (your item 2), in place of the extension discussed in item 10. Proposition 4.10, new: Γ_{g*} is divisible for an interior g* if and only if Γ_0 = (Γ_a − gΓ_ref)/(1 − g) for a divisible Γ_a with Γ_a ≥ gΓ_ref entrywise; then X = gK at every triple and D_CK(δt;Γ_g) = (1 − g)|g* − g| M_(δt). The instance is your Glauber pair (reference λ = 2, extrapolant λ = 1, g = 0.3603), with the constants in closed form: with a(τ) = (e^{−2τ} − e^{−4τ})/(1 − g*), M_(δt) = a(½ ln 2)² = 1/[16(1 − g)²] = 0.152731 for δt ≥ ln 2, D_CK(δt;Γ_0) = gM_ = 0.055029, and the rebound peak is (1 − g*)²M_/4 = 1/64 = 0.015625 exactly, at g = (1 + g)/2. At ε = 0.01: g_first = 0.2705, g_term = 0.8721, g_1 = 0.9500, g_0 = 0.9519, as you report. One figure differs: the re-entry point from the closed form is 0.4882, not 0.4885 (g² − (1 + g*)g + g* + ε/M_* = 0 has roots 0.4882 and 0.8721), and the manuscript prints 0.4882. The abstract now says that a two-state family exhibits the interior zero, the rebound, and the threshold separation at the scale level; the reference to the Koopman paper's Section 6 family is removed. Sara, the pointwise example keeps the status statement you asked for; the scale-level family sits after it.
>
> Wording. (i) Abstract: "exact vanishing is guaranteed at g = 1 and possible at no more than one interior coupling." (ii) After Proposition 6.1: "reduces to the bound (7) of Theorem 3.1." (iii) Remark 6.2 states the weight for states of positive probability under the mixture; a null column is completed with the constant g, which keeps the family in class (16) and leaves δ_g and the estimate unchanged, because both one-time laws sum to one over the states of positive probability, so g lies between the smallest and the largest Bayes weight at each s. (iv) Proof of Theorem 4.3(i): the radicand is positive because (D + M)² − 4Mε > (D + M)² − 4MD = (D − M)² ≥ 0 for ε < D.
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> Proposition 4.10 is the only new numbered item; every other number is unchanged, so the references in both of your letters still hold.
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> Let me know your thoughts.
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> Ben
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> Benjamin D. Summers
> *Managing Director*
> **ADAGIO GROUP**
> **MOBILE** +1.713.775.1011 | **OFFICE** +1.305.433.1823 | **FAX** +1 239.932.5558
> bsummers@adagiollc.com | adagio.capital

On Wed, Sep 23, 2026 at 5:34 PM Ezequiel Boero <ezequiel.boero@adagioinstitute.org> wrote: